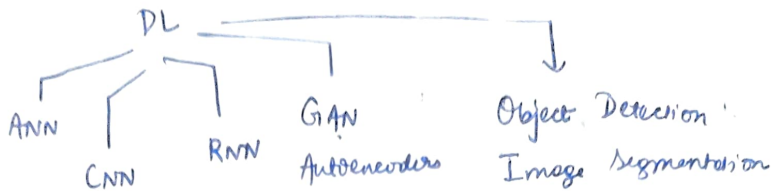


Curriculum

1. Artificial Neural Network

a. Basis

b. Perception

c. MLP

d. Training an MLP

e. Practical with Keras.

f. How to improve an ANN

g. Advanced Topics.

h. Project.

Features:

4) TensorFlow + Keras

5) Projects.

Prerequisites

1) Python ✓

2) Basis of ML ✓

3) Linear Algebra ✓ (3 Blue 1 Brown)

Extra Content

1) Deep Learning Roadmap

2) DL Project Ideas

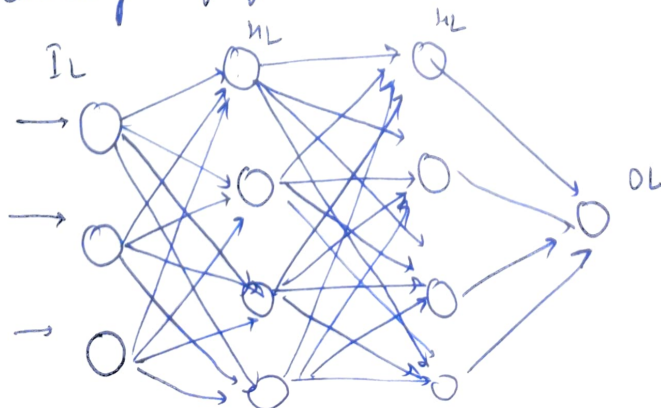
3) Projects

4) Interview Questions (IQ).

②. What is Deep Learning?

→ DL is a subfield of AI & ML i.e., inspired by structure of human brain

→ DL algorithms attempt to draw similar conclusions as humans would by continually analyzing data with given logical structure called NN.



Why DL is getting famous?

- 1) Applicability → medical, defence,
- 2) Performance → state-of-the-art result

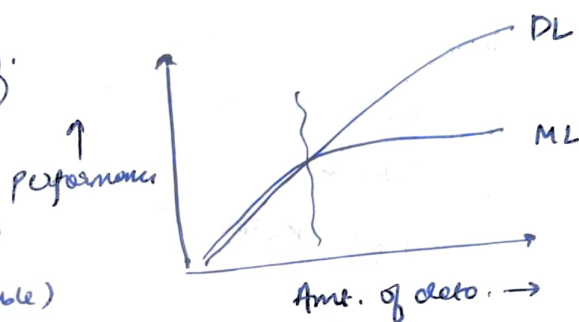
DL → part of → ML methods based on ANN with representation learning.

DL uses multiple layers to progressively extract higher level features from raw input.

For e.g., In image processing — lower layers → edges
higher layers → concepts relevant to human such as digits or letters.

Deep Learning vs Machine Learning

1. Data Dependency (high)
2. Hardware dependency → (GPU) ^{costly}
3. Training time (high)
4. Feature selection (automatic)
5. Interpretability (not interpretable)



Prediction time → fast

Why Now? BT

- + Data sets
- + Frameworks
- + Architecture
- + Hardware
- + Community

2010+ → smart phones
Internet pricing

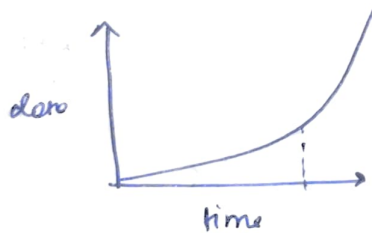


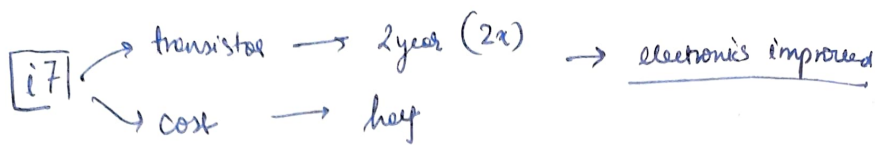
image classification → dog
→ cat

unlabelled data
↓
Create labels.

Public datasets :-

1. Microsoft COCO — Image
2. Youtube 8M — 6.1 million videos
3. SQuAD — Wikipedia — 150,000 QA
4. Audio → Google AudioSet → 20,000,000 audio samples.

Hardware



• FPGA → Field Programmable Gate Array → fast, low power, reprog → custom

• NVIDIA

• ASIC → Application Specific Integrated Circuit

- + TPU (Tensor Processing Unit)
- + Edge TPU → Drone, watch, Smart-glass.
- + Neural Processing Units (NPU)

Factor #3 → Frameworks / Libraries

1. TensorFlow by Google → Difficult to use
2. PyTorch by Facebook
3. Keras → Frontend for ~~Keras~~ TF

Facebook

2016 → PyTorch

AI Research

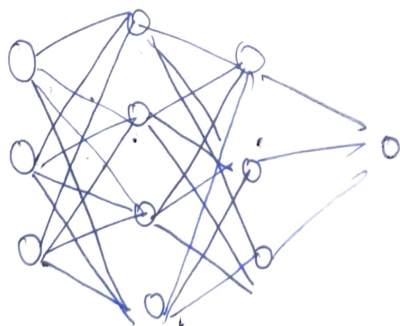
(2013) → Caffe2

PyTorch → TensorFlow

① AutoML

② CustomVision42

Factor #4 → Deep Learning Architectures



Experiment

- + time
- + effort
- money

So, we use existing architectures → Transfer Learning

image classification → ResNET

Text classification → BERT

image segmentation → UNet

image translation → Pix2Pix

Object detection → YOLO

Speech generation → WordNET

Factor #5 → People / Community